

Linear algebra – 3

Eigenvalues, diagonalisability and the Jordan form

12 сентября 2026 года

Theoretical minimum.

Definition. The minimal polynomial μ_A is the monic generator of the ideal $\{f : f(A) = O\}$; it divides χ_A and has the same roots. A is diagonalisable over a field K if and only if μ_A splits over K and has no repeated roots.

Theorem 1 (Jordan form). Every complex matrix is similar to a direct sum of Jordan blocks, uniquely up to the order of the blocks; the number of blocks of size $\geq k$ with eigenvalue λ equals $\text{rank}(A - \lambda E)^{k-1} - \text{rank}(A - \lambda E)^k$. Consequently the ranks of the powers of $A - \lambda E$ determine the whole picture.

Theorem 2 (Frobenius normal form). Over any field A is similar to a direct sum of companion matrices of polynomials $f_1 \mid f_2 \mid \cdots \mid f_k$, with $f_k = \mu_A$ and $f_1 \cdots f_k = \chi_A$. In particular similarity over a field is not changed by passing to a larger field: two real matrices similar over $\mathbb{C}$ are similar over $\mathbb{R}$.

Theorem 3 (the spectral theorem and what it gives). A real symmetric matrix has an orthonormal eigenbasis and real eigenvalues; $\text{tr } A = \sum \lambda_i$, $\det A = \prod \lambda_i$, and in general the coefficients of χ_A are the elementary symmetric functions of the eigenvalues, so that $\sum_{i < j} \lambda_i \lambda_j$ is the sum of the principal 2×2 minors.

Fact 1 (the spectral mapping principle). If λ is an eigenvalue of A then $f(\lambda)$ is an eigenvalue of $f(A)$ for every polynomial f , and over $\mathbb{C}$ every eigenvalue of $f(A)$ arises this way. Hence a polynomial identity $f(A) = O$ confines the spectrum to the roots of f , and if those roots are simple the matrix is diagonalisable — this single remark solves a whole genre of problems.

Fact 2 (commuting matrices). Commuting matrices have a common eigenvector and hence can be simultaneously triangularised over $\mathbb{C}$, but they need not be simultaneously reducible to Jordan form.

1. Prove that A is diagonalisable if and only if μ_A has simple roots, construct the Frobenius normal form, and deduce that two matrices are similar over a field K if and only if they are similar over any extension of K .
2. (VJIMC 2016) For $n \geq 3$ find the eigenvalues, with their multiplicities, of the $n \times n$ matrix A with $a_{11} = a_{nn} = 1$, $a_{ii} = 2$ for $1 < i < n$, $a_{i,i+2} = a_{i+2,i} = 1$ for all admissible i , and all other entries zero. For $n = 7$ it looks like

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

3. (ISTCiM 2023) Let A be a complex $d \times d$ matrix and assume that there exists a positive integer n with $A^{2n} + A^n + 2nE = O$. Prove that A is diagonalisable.
4. (IMC 2013) Let A and B be real symmetric matrices with all eigenvalues strictly greater than 1. Let λ be a real eigenvalue of AB . Prove that $|\lambda| > 1$.
5. (IMC 2014) Let $A = (a_{ij})_{i,j=1}^n$ be a symmetric $n \times n$ matrix with real entries and let $\lambda_1, \dots, \lambda_n$ be its eigenvalues. Show that

$$\sum_{1 \leq i < j \leq n} a_{ii}a_{jj} \geq \sum_{1 \leq i < j \leq n} \lambda_i \lambda_j,$$

and determine all matrices for which equality holds.

6. (VJIMC 2024) A real 2024×2024 matrix A is called *nice* if $\langle Av, v \rangle = 1$ for every vector $v \in \mathbb{R}^{2024}$ of unit norm.
 - a) Prove that the only nice matrix all of whose eigenvalues are real is the identity matrix.
 - b) Find an example of a nice matrix different from the identity.
7. (MexMAT, 2006) Do there exist two commuting linear operators on $\mathbb{R}^3$ such that there is no basis in which both matrices are in Jordan form?
8. (SEEMOUS 2013) Let $M, N \in M_2(\mathbb{C})$ be non-zero with $M^2 = N^2 = O_2$ and $MN + NM = E_2$. Prove that there is an invertible $A \in M_2(\mathbb{C})$ with $M = A \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} A^{-1}$ and $N = A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} A^{-1}$.